

HW 5 homework debrief

Supp IV.

$A + B$ are $n \times n$ complex matrices that commute: $AB = BA$. WTS A and B share at least one eigenvector in common. It's enough to show that they have a common eigen"line".

Since they are complex operators, they each have at least one eigenvector. ~~We want to show that for instance, B cannot move an eigenvector of A off of its line.~~ Let v be an eigenvector of A with eigenvalue λ .

Since $BAv = B\lambda v = \lambda Bv = \lambda w$, where w is the image of v under B .

Then $ABv = Aw$

Then since $BA = AB$, we have $Aw = \lambda w$

So w is also an eigenvector of A , also with eigenvalue λ .

What does B do to w ? If it keeps it in the span of $\{v, w\}$, then B has an eigenvector in that plane (since it is an invariant plane under B) and so $A + B$ share an eigenvector, since $\text{span}(\{v, w\})$ is (contained in) the λ -eigenspace of A .

But what if B moves w out of $\text{span}(\{v, w\})$?
Maybe to some vector x ? Then we again have
(by a similar calculation as above)

$$BAw = B\lambda w = \lambda Bw = \lambda x$$

$$ABw = Ax$$

and so $Ax = \lambda x$ and x is also in
the λ -eigenspace of A !

Iterating on this, since n is finite,
eventually B must have an eigenvector in
the λ -eigenspace of A . Because it either
has an invariant subspace within the λ -eigenspace
of A ; ~~or the whole~~ or we arrive at
the stage where all of $\mathbb{C}^n$ is the λ -eigenspace
of A , and so A shares an eigenvector
with B , since B has an eigenvector
and every non-zero vector in $\mathbb{C}^n$ is
an eigenvector of A .

4.10

Prove that the determinant of a matrix A is the product of its eigenvalues (counting multiplicities)

The eigenvalues of A are the roots of the characteristic polynomial of A : $\det(A - \lambda I)$.

For A an $n \times n$ complex matrix, this is a degree n ^{complex} polynomial; by the fundamental theorem of ~~linear~~ algebra, this has n roots (counting multiplicity) and so factors as a product of n linear factors:

$$\det(A - \lambda I) = (\lambda_1 - \lambda)(\lambda_2 - \lambda) \dots (\lambda_n - \lambda)$$

Now, if we let $\lambda = 0$ on both sides of this equation, we get:

$$\det(A) = (\lambda_1)(\lambda_2) \dots (\lambda_n)$$

That is, $\det(A)$ equals the product of the eigenvalues of A .